



Modelling Financial Markets

Seminar 3: Factor Models and Asset Pricing — Hands-on



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Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CMA: Conservative Minus Aggressive (investment factor)
- ▣ ETF: Exchange-Traded Fund
- ▣ FDR: False Discovery Rate
- ▣ FF3: Fama–French three-factor model
- ▣ FF5: Fama–French five-factor model
- ▣ FWER: Family-Wise Error Rate
- ▣ GLS: Generalised Least Squares
- ▣ GMM: Generalized Method of Moments
- ▣ GRS: Gibbons–Ross–Shanken (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HML: High Minus Low (value factor)
- ▣ KRS: Kan–Robotti–Shanken (standard errors)
- ▣ MOM: MOMentum factor
- ▣ NW: Newey–West (standard errors)
- ▣ OLS: Ordinary Least Squares
- ▣ PC1: first Principal Component
- ▣ PC2: second Principal Component
- ▣ PC3: third Principal Component
- ▣ PCA: Principal Component Analysis
- ▣ RF: Risk-Free rate
- ▣ RMW: Robust Minus Weak (profitability factor)

Acronyms used in this material (2/2)

- **SMB: Small Minus Big (size factor)**
- **US: United States**

Outline

Introduction

Key Definitions

Part A: Derivations

Part B: Estimation and Inference

Part C: Open Question

Wrap-up

Seminar Objectives

By the end of this seminar you can

- ▣ Derive the CAPM, the Sharpe-ratio form of GRS and the Vasicek posterior mean
- ▣ Estimate betas and alphas with robust and bootstrap inference
- ▣ Test the CAPM jointly (GRS) and estimate risk premia (Fama–MacBeth) with misspecification-robust errors
- ▣ Compare factor models and choose the number of latent factors

- ▣ Background: Lecture 3; textbook: [Campbell, Lo & MacKinlay \(1997\)](#), ch. 5–6
- ▣ Seminar: Drd. Antoaneta Amza, antoaneta.amza@csie.ase.ro

Structure

- ▣ **Part A:** eight derivations (A1–A8)
- ▣ **Part B:** eleven computational exercises with inference, US and Romanian data (B1–B11)
- ▣ **Part C:** one open question, a starting point for the team project (C1), and an AI answer to critique (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every Part B exercise ends with an **interpretation question**
 - ▶ It cannot be answered by reading a number: ask what would change the conclusion
- Report every estimate with its uncertainty (standard error, interval or p -value)
- Before any regression: align the calendars and look for data errors (unadjusted corporate events)

Solved and Proposed Problems

Solved: worked examples in this deck

- ▣ Part A: A1 (CAPM from the tangency portfolio), A3 (Vasicek as a posterior mean), A5 (Bonferroni, Holm, BH, step by step), A7 (errors in variables in pass 2)
- ▣ Part B: B1 (beta and alpha of XLK with HAC and bootstrap), B2 (GRS on sector ETFs), B3 (PCA and the number of factors)

Proposed: you solve them

- ▣ Part A: A2 (model: A1); A4 (model: A3); A6 (model: A7); A8 (model: B3)
- ▣ Part B: B4, B6, B10 (model: B1); B5, B9 (model: B2 and Lecture 3, Section 5); B7 (model: A5); B8 (model: B3); B11 (model: B2)
- ▣ Part C: C1, open question (model: B1, B5); C2, critique an AI answer (model: C1)

Setup: Python Environment and Data

-  **Open in Colab:** [chapter3_seminar_notebook.ipynb](#)
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `matplotlib`

```
import numpy as np, pandas as pd
URL = "https://raw.githubusercontent.com/danpele/MFM/main/data/market/"
x1k = pd.read_csv(URL + "XLK.US.csv", index_col="date",
                  parse_dates=True)["adjusted_close"]
FF = "https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/"
# F-F_Research_Data_5_Factors_2x3_CSV.zip
```

- ETFs and stocks: `adjusted_close`; BET: `close`; factors: [Kenneth French Data Library](#) (percent, divide by 100)
- Joint analyses: join the *prices* on common days first, then take returns

Key Definitions: CAPM and Beta

- ▣ **Excess return** $R_{i,t}^e = R_{i,t} - R_{f,t}$; **CAPM** (Sharpe (1964)): $\mathbb{E}[R_i^e] = \beta_i \mathbb{E}[R_m^e]$
- ▣ **Beta** $\beta_i = \text{Cov}(R_i, R_m) / \text{Var}(R_m)$; **Jensen's alpha**: intercept of $R_{i,t}^e = \alpha_i + \beta_i R_{m,t}^e + \varepsilon_{i,t}$
- ▣ **Portfolio beta** $\beta_p = \sum_i w_i \beta_i$ (betas are linear in the weights)
- ▣ **Blume** (Blume (1971)): $\beta^B = 0.33 + 0.67 \hat{\beta}$
- ▣ **Vasicek** (Vasicek (1973)): $\beta^V = w \hat{\beta} + (1 - w) \bar{\beta}$, $w = \sigma_\beta^2 / (\sigma_\beta^2 + \text{se}(\hat{\beta})^2)$
- ▣ **HAC** (Heteroskedasticity and Autocorrelation Consistent) standard errors, Newey & West (1987): Bartlett weights $1 - l / (L + 1)$

Key Definitions: Tests and Latent Factors

- ▣ **GRS** (Gibbons, Ross & Shanken (1989)): $W = \frac{T-N-1}{N} \frac{\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \hat{\mu}_m^2 / \hat{\sigma}_m^2} \sim F(N, T - N - 1)$ under H_0 : all $\alpha_i = 0$
- ▣ **Fama–MacBeth** (Fama & MacBeth (1973)): monthly cross-sectional slopes $\hat{\lambda}_t$, averaged; **Shanken** (Shanken (1992)) scales the non-factor part of their variance by $(1 + c)$, $c = \hat{\lambda}^\top \Sigma_f^{-1} \hat{\lambda}$:

$$\text{Var}_{\text{Sh}} = (1 + c)(\text{Var}_{FM} - \Sigma_f / T) + \Sigma_f / T$$
- ▣ **Kan–Robotti–Shanken** (Kan, Robotti & Shanken (2013)): both passes as one GMM system; errors robust to a misspecified model
- ▣ **Bonferroni**: $p_j \leq \alpha / M$; **Holm** (Holm (1979)): $p_{(j)} \leq \alpha / (M - j + 1)$ step by step; **BH** (Benjamini & Hochberg (1995)): largest j with $p_{(j)} \leq j\alpha / M$
- ▣ **PCA** (Principal Component Analysis): eigenvalues λ_k and eigenvectors of the correlation matrix; share λ_k / N ; number of factors: Bai & Ng (2002), Ahn & Horenstein (2013)
- ▣ **Moving-block bootstrap** (Künsch (1989)): resample blocks of consecutive observations, keeping the dependence inside each block

Exercise A1: The CAPM from the Tangency Portfolio [Solved]

Task

- ▣ **Setting:** N risky assets with expected excess returns $\boldsymbol{\mu}$ and positive definite covariance matrix $\boldsymbol{\Sigma}$, plus a risk-free asset
- ▣ **Assumption:** the tangency portfolio $\mathbf{w}_m = \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}/(\mathbf{1}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu})$ is the market portfolio, $R_m = \mathbf{w}_m^\top \mathbf{R}$
- ▣ (a) Express the covariance vector $\text{Cov}(\mathbf{R}, R_m) = \boldsymbol{\Sigma}\mathbf{w}_m$ as a function of $\boldsymbol{\mu}$
- ▣ (b) Compute $\mu_m = \mathbf{w}_m^\top \boldsymbol{\mu}$ and $\sigma_m^2 = \mathbf{w}_m^\top \boldsymbol{\Sigma} \mathbf{w}_m$ and find the relation between them
- ▣ (c) Combine (a) and (b) to prove $\mu_i = \beta_i \mu_m$ for every asset, with $\beta_i = \text{Cov}(R_i, R_m)/\sigma_m^2$
- ▣ (d) Explain why the same result holds for any mean–variance efficient portfolio
- ▣ **Report:** the derivation in three steps and one sentence for (d)

Exercise A1: Solution [Solved]

Solution

- Step 1: let $c = 1^\top \Sigma^{-1} \mu$, a scalar. Then $\Sigma w_m = \Sigma \Sigma^{-1} \mu / c = \mu / c$, so $\text{Cov}(R, R_m) = \mu / c$
- Step 2: $\sigma_m^2 = w_m^\top \Sigma w_m = w_m^\top \mu / c = \mu_m / c$, hence $c = \mu_m / \sigma_m^2$
- Step 3: $\mu = c \text{Cov}(R, R_m) = \mu_m \text{Cov}(R, R_m) / \sigma_m^2 = \beta \mu_m$: the security market line
- (d) Steps 1–3 used only $w_m \propto \Sigma^{-1} \mu$, that is, efficiency. Any efficient portfolio prices all assets exactly by its betas, so a CAPM test is also a test of the efficiency of the market proxy (Roll's critique)

Exercise A2: GRS Equals a Sharpe-Ratio Gain [Proposed]

Solution discussed in class

Task

- ▣ **Setting:** one factor f (the market excess return) and N test assets R ; maximum-likelihood moments $\mu_f, \sigma_f^2, \mu_R, \Sigma_R, \sigma_{Rf}$
- ▣ **Definitions:** $\beta = \sigma_{Rf}/\sigma_f^2$, $\alpha = \mu_R - \beta\mu_f$, residual covariance $\Sigma = \Sigma_R - \beta\beta^\top \sigma_f^2$
- ▣ (a) Write $V = \text{Cov}((f, R^\top)^\top)$ in partitioned form and invert it with the Schur complement of σ_f^2
- ▣ (b) Show that $SR^2(f, R) = \mu_Y^\top V^{-1} \mu_Y$ equals $\mu_f^2/\sigma_f^2 + \alpha^\top \Sigma^{-1} \alpha$, so that $\alpha^\top \Sigma^{-1} \alpha = SR^2(f, R) - SR^2(f)$
- ▣ (c) Write the GRS statistic W as a function of the two Sharpe ratios
- ▣ (d) Compute W for the 25 size $\times$ B/M portfolios: $T = 757$, $N = 25$, monthly $SR(f) = 0.134$, $SR(f, R) = 0.405$; the 5% critical value of $F(25, 731)$ is 1.52
- ▣ **Report:** the derivation, W , the decision at 5% and one sentence on the economic meaning. **Model:** A1



Exercise A3: Vasicek as a Posterior Mean [Solved]

Task

- ▣ **Data:** 9 sector ETFs (XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY), monthly excess returns on SPY, January 1999 – December 2012; OLS betas with Newey–West errors
- ▣ **Numbers:** XLK $\hat{\beta} = 1.415$, $\text{se} = 0.139$; cross-section: mean $\bar{\beta} = 0.949$, variance of the estimated betas 0.116, average squared standard error 0.0077; realised XLK beta in 2013–2026: 1.172
- ▣ **Assumptions:** prior $\beta_i \sim N(\bar{\beta}, \sigma_{\beta}^2)$; likelihood $\hat{\beta}_i | \beta_i \sim N(\beta_i, \text{se}_i^2)$ with se_i known
- ▣ (a) Derive the posterior distribution of β_i given $\hat{\beta}_i$ and its mean β_i^V
- ▣ (b) Use the law of total variance to show $\text{Var}_{cs}(\hat{\beta}) = \sigma_{\beta}^2 + \mathbb{E}[\text{se}^2]$, and write the empirical-Bayes estimator of σ_{β}^2
- ▣ (c) Compute σ_{β}^2 , the weight w and β^V for XLK, and the Blume beta $0.33 + 0.67\hat{\beta}$
- ▣ (d) Compare the absolute errors of the raw, Vasicek and Blume betas against the realised 1.172
- ▣ **Report:** the formulas, the four numbers of (c) and the three errors of (d)

Exercise A3: Solution (1/2) [Solved]

Solution (1/2)

- (a) Posterior density $\propto \exp\left\{-\frac{(\beta - \bar{\beta})^2}{2\sigma_{\beta}^2}\right\} \exp\left\{-\frac{(\hat{\beta} - \beta)^2}{2\text{se}^2}\right\}$: the exponent is quadratic in β , so the posterior is Normal
- Its precision is $1/\sigma_{\beta}^2 + 1/\text{se}^2$ and its mean is $\frac{\bar{\beta}/\sigma_{\beta}^2 + \hat{\beta}/\text{se}^2}{1/\sigma_{\beta}^2 + 1/\text{se}^2} = w\hat{\beta} + (1 - w)\bar{\beta}$, with $w = \frac{\sigma_{\beta}^2}{\sigma_{\beta}^2 + \text{se}^2}$
- (b) $\text{Var}(\hat{\beta}) = \text{Var}(\mathbb{E}[\hat{\beta} \mid \beta]) + \mathbb{E}[\text{Var}(\hat{\beta} \mid \beta)] = \text{Var}(\beta) + \mathbb{E}[\text{se}^2]$
- Hence $\hat{\sigma}_{\beta}^2 = \text{Var}_{cs}(\hat{\beta}) - \overline{\text{se}^2}$; using $\text{Var}_{cs}(\hat{\beta})$ itself counts the sampling noise as true dispersion and shrinks too little

Exercise A3: Solution (2/2) [Solved]

Solution (2/2)

- ▣ (c) $\sigma_{\beta}^2 = 0.116 - 0.008 = 0.108$ and $se^2 = 0.139^2 = 0.0193$
- ▣ $w = 0.108 / (0.108 + 0.0193) = 0.849$, so $\beta^V = 0.849 \times 1.415 + 0.151 \times 0.949 = 1.344$
- ▣ With the raw variance 0.116 the weight would be 0.857 and $\beta^V = 1.348$; Blume: $0.33 + 0.67 \times 1.415 = 1.278$
- ▣ (d) Errors against 1.172: raw 0.243, Vasicek 0.172, Blume 0.106
- ▣ Interpretation: XLK's beta is estimated precisely, so Vasicek shrinks it only a little; Blume shrinks every beta by a third and wins here. Exercise A4 shows why regression to the mean exceeds sampling noise

Exercise A4: Blume's Slope as a Shrinkage Weight [Proposed]

Solution discussed in class

Task

- ▣ **Data:** betas of the same 9 sector ETFs in two non-overlapping samples, 1999–2012 ($\hat{\beta}_{i,1}$) and 2013–2026 ($\hat{\beta}_{i,2}$); fitted cross-sectional line $\hat{\beta}_2 = 0.230 + 0.724 \hat{\beta}_1$; from A3: $\sigma_{\beta}^2 = 0.108$, $\overline{se^2} = 0.0077$, $\bar{\beta} = 0.949$
- ▣ **Assumptions:** true betas are constant, $\hat{\beta}_{i,k} = \beta_i + \eta_{i,k}$, the noises are independent of β_i and of each other, with common variance se^2
- ▣ (a) Derive the probability limit, as $N \rightarrow \infty$, of the OLS slope b of $\hat{\beta}_2$ on $\hat{\beta}_1$ and of the intercept
- ▣ (b) Show that b equals the Vasicek weight with a common standard error, so that Blume's rule is a Vasicek estimator
- ▣ (c) Compute the implied slope and intercept from the A3 numbers and compare them with 0.724 and 0.230
- ▣ (d) State which assumption must fail to explain the gap
- ▣ **Report:** the probability limits, the implied numbers and a one-sentence interpretation. **Model:** A3

Exercise A5: Ten Candidate Factors [Solved]

Task

- ▣ **Setting:** ten new candidate factors, each tested separately with a two-sided t -test of zero mean; level $\alpha = 5\%$, $M = 10$
- ▣ **Sorted p -values:** 0.001, 0.004, 0.009, 0.012, 0.021, 0.035, 0.048, 0.060, 0.20, 0.55
- ▣ (a) Count the discoveries of the naive rule $p \leq \alpha$
- ▣ (b) Apply Bonferroni: reject H_j if $p_j \leq \alpha/M$
- ▣ (c) Apply Holm step by step: compare $p_{(j)}$ with $\alpha/(M - j + 1)$ and stop at the first failure
- ▣ (d) Apply Benjamini–Hochberg: find the largest j with $p_{(j)} \leq j\alpha/M$ and reject $H_{(1)}, \dots, H_{(j)}$
- ▣ (e) State which error rate each procedure controls, and under which dependence between the tests
- ▣ **Report:** the four counts and one sentence on which procedure suits a factor search

Exercise A5: Solution [Solved]

Solution

- (a) Seven p -values are at most 0.05: **7** discoveries
- (b) $\alpha/M = 0.005$: only 0.001 and 0.004 pass: **2**
- (c) Holm thresholds 0.005, 0.0056, 0.00625: $0.001 \leq 0.005$, $0.004 \leq 0.0056$, $0.009 > 0.00625$, stop: **2**
- (d) BH thresholds $0.005j$: 0.005, 0.010, 0.015, 0.020, 0.025, 0.030, $\dots$; $p_{(5)} = 0.021 \leq 0.025$, while $p_{(6)} = 0.035 > 0.030$ and no later $p_{(j)}$ passes: **5**
- (e) Bonferroni and Holm control the FWER under any dependence; BH controls the FDR for independent or positively dependent tests
- Interpretation: in a factor search the share of false factors (FDR) is the relevant target; among FWER methods Holm always rejects at least as much as Bonferroni

Exercise A6: A Useless Factor in Pass 2 [Proposed]

Solution discussed in class

Task

- **Setting:** N test assets with excess returns R_{it} , $t = 1, \dots, T$, and means μ_i ; a candidate factor g_t , i.i.d., with mean 0 and variance σ_g^2 , independent of all returns, so $\beta_{i,g} = 0$
- (a) Show that the OLS slope $\hat{\beta}_{i,g} = \widehat{\text{Cov}}(R_i, g) / \hat{\sigma}_g^2$ satisfies $\sqrt{T} \hat{\beta}_{i,g} \rightarrow N(0, \sigma_i^2 / \sigma_g^2)$, so $\hat{\beta}_{i,g} = O_p(T^{-1/2})$
- (b) With g alone in pass 2, write $\hat{\lambda}_g = \widehat{\text{Cov}}_{cs}(\bar{R}_i, \hat{\beta}_{i,g}) / \widehat{\text{Var}}_{cs}(\hat{\beta}_{i,g})$ in terms of $z = \sqrt{T} \hat{\beta}_g$ and show that $\hat{\lambda}_g = O_p(\sqrt{T})$ when the μ_i differ across assets
- (c) Show that when all μ_i are equal, $\hat{\lambda}_g$ still has a non-degenerate random limit, not 0
- (d) Explain why the Fama–MacBeth t -test of $\lambda_g = 0$ over-rejects (Kan & Zhang (1999)) and which first-pass test protects against it
- **Report:** the orders of magnitude, the recommended check and a comparison with the Lecture 3 simulation. **Model:** A7

Exercise A7: Errors in Variables in Pass 2 [Solved]

Task

- ▣ **Setting:** one-factor model $R_{it} = a_i + \beta_i f_t + \varepsilon_{it}$, with ε_{it} i.i.d. over time, common variance σ_ε^2 , independent of f ; pass 2 is the OLS regression of $\bar{R}_i$ on $(1, \hat{\beta}_i)$
- ▣ **Noise:** $\hat{\beta}_i = \beta_i + \eta_i$ with $\eta_i = \sum_t (f_t - \bar{f}) \varepsilon_{it} / \sum_t (f_t - \bar{f})^2$
- ▣ (a) Show that $\text{Var}(\eta_i | f) = \sigma_\varepsilon^2 / (T \hat{\sigma}_f^2)$
- ▣ (b) Show that η_i is uncorrelated with $\bar{\varepsilon}_i$ and that, for $N \rightarrow \infty$ and T fixed, the pass-2 slope converges to $\lambda \text{Var}_{cs}(\beta) / [\text{Var}_{cs}(\beta) + \sigma_\varepsilon^2 / (T \sigma_f^2)]$
- ▣ (c) Estimate this attenuation factor for the CAPM on the 25 size $\times$ B/M portfolios (monthly 1963–2026, $T = 757$) and on the 9 sector ETFs (1999–2026, $T = 331$), with $\text{Var}_{cs}(\beta) = \text{Var}_{cs}(\hat{\beta}) - \bar{\sigma}_\varepsilon^2 / (T \hat{\sigma}_f^2)$
- ▣ (d) Explain what the Shanken correction fixes and what it does not
- ▣ **Report:** the derivation, the two attenuation factors and one sentence on single stocks

Exercise A7: Solution (1/2) [Solved]

Solution (1/2)

- (a) Given f , η_i is a linear combination of the ε_{it} with weights $(f_t - \bar{f}) / \sum_s (f_s - \bar{f})^2$, so

$$\text{Var}(\eta_i | f) = \sigma_\varepsilon^2 / \sum_t (f_t - \bar{f})^2 = \sigma_\varepsilon^2 / (T \hat{\sigma}_f^2)$$
- (b) $\text{Cov}(\bar{\varepsilon}_i, \eta_i | f) = \sigma_\varepsilon^2 \frac{1}{T} \sum_t (f_t - \bar{f}) / \sum_s (f_s - \bar{f})^2 = 0$, because $\sum_t (f_t - \bar{f}) = 0$
- Write $\bar{R}_i = \lambda \mathbf{0} + \lambda \beta_i + u_i$ with u_i uncorrelated with η_i ; then $\widehat{\text{Cov}}_{cs}(\bar{R}, \hat{\beta}) \rightarrow \lambda \text{Var}_{cs}(\beta)$ and $\widehat{\text{Var}}_{cs}(\hat{\beta}) \rightarrow \text{Var}_{cs}(\beta) + \sigma_\varepsilon^2 / (T \sigma_f^2)$
- The ratio is the stated limit: the classical errors-in-variables attenuation towards zero

Exercise A7: Solution (2/2) [Solved]

Solution (2/2)

- (c) 25 portfolios: $\text{Var}_{cs}(\hat{\beta}) = 0.0191$, noise $\bar{\sigma}_\varepsilon^2 / (T \hat{\sigma}_f^2) = 0.0006$, $\text{Var}_{cs}(\beta) = 0.0184$; factor $0.0184/0.0191 = 0.967$
- 9 sectors: $\text{Var}_{cs}(\hat{\beta}) = 0.0925$, noise 0.0020 , $\text{Var}_{cs}(\beta) = 0.0905$; factor 0.978
- Interpretation: the bias is only 2–3% here, because diversified portfolios have small residual variance and T is long; single stocks have much larger σ_ε^2 , so the attenuation grows
- (d) [Shanken \(1992\)](#) corrects the standard error of $\hat{\lambda}$ for the noise in $\hat{\beta}$ (a variance correction); it does not remove the finite- T bias of the point estimate

Exercise A8: Equicorrelated Assets [Proposed]

Solution discussed in class

Task

- ▣ **Setting:** N standardised returns with equal pairwise correlation $\rho = 0.5$; correlation matrix $C = (1 - \rho)I + \rho \mathbf{1}\mathbf{1}^\top$
- ▣ (a) For $N = 3$, find the eigenvalues and eigenvectors of C
- ▣ (b) Compute the share of total variance explained by the first component, λ_1/N
- ▣ (c) Generalise to N assets and find the limit of the PC1 share as $N \rightarrow \infty$
- ▣ (d) Invert the formula of (c) to obtain the average correlation implied by a PC1 share s ; apply it to the BVB blue chips ($N = 10$, $s = 0.425$, B8) and to the US sectors ($N = 11$, $s = 0.660$, B3)
- ▣ **Report:** eigenvalues, the PC1 share, the limit and the two implied correlations. **Model:** B3 and the PCA recap of Lecture 3 (two assets: $\lambda = 1 \pm \rho$)

Exercise B1: Beta and Alpha of XLK [Solved]

Task

- **Data:** XLK (technology ETF) adjusted prices, file `XLK.US.csv` in `data/market`, month-end values; Mkt-RF and RF from the [Kenneth French Data Library](#); monthly excess returns January 1999 – July 2026 ($T = 331$)
- (a) Estimate $R_{\text{XLK},t}^e = \alpha + \beta \text{Mkt-RF}_t + \varepsilon_t$ by OLS and annualise α (multiply by 12)
- (b) Compute classical and Newey–West standard errors of α and β , with Bartlett weights and $L = \lfloor 4(T/100)^{2/9} \rfloor$ lags
- (c) Build a 95% moving-block bootstrap interval for α : blocks of 12 months, 2000 draws, percentile method ([Künsch \(1989\)](#))
- (d) Test $H_0 : \alpha = 0$ at 5% with the Newey–West t -statistic
- **Report:** a table with estimates, classical and Newey–West errors, $t(\alpha)$, R^2 and the bootstrap interval

Interpretation

- The HAC error of beta is twice the classical one: which feature of the residuals causes it?

Exercise B1: Solution (1/2) [Solved]

Solution (1/2)

- (b) Lags: $L = \lfloor 4 \times 3.31^{2/9} \rfloor = \lfloor 5.2 \rfloor = 5$

	Estimate	Classical s.e.	Newey–West s.e.
α (annualised)	0.33%	2.27%	2.45%
β	1.277	0.041	0.082

- (a) $R^2 = 0.742$; the robust 95% interval of β is $1.277 \pm 1.96 \times 0.082 = [1.12, 1.44]$ instead of $[1.20, 1.36]$
- (d) $t(\alpha) = 0.33/2.45 = 0.13$, $p = 0.89 > 0.05$: do not reject; no evidence of a CAPM alpha
- (c) Bootstrap: $\alpha \in [-5.4\%, 4.8\%]$, standard error 2.61%; the interval contains 0, as the Newey–West test says

Exercise B1: Solution (2/2) [Solved]

Solution (2/2)

- Interpretation: classical errors assume $\text{Var}(\varepsilon_t \mid f_t)$ is constant
- The sandwich estimator weights each squared residual by $(f_t - \bar{f})^2$: residual variance is high exactly when the market moves a lot (tech bust 2000–2002, 2008, 2020)
- This heteroskedasticity makes the classical formula understate $\text{Var}(\hat{\beta})$: the ratio of errors is $0.082/0.041 = 2.0$
- The point estimates are the same; only the uncertainty changes

Exercise B2: GRS Test on Sector ETFs [Solved]

Task

- ▣ **Data:** 9 sector ETFs (XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY), month-end adjusted prices joined on common months; monthly excess returns January 1999 – July 2026 ($T = 331$); market factor Mkt-RF
- ▣ (a) Estimate the 9 time-series regressions on Mkt-RF and report the annualised alphas and the betas
- ▣ (b) Compute the GRS statistic with the maximum-likelihood residual covariance and its p -value from $F(9, 321)$
- ▣ (c) Compute the monthly Sharpe ratio of the market and of the tangency portfolio of market plus sectors, and check the identity of A2
- ▣ (d) Decide at the 5% level
- ▣ **Report:** alphas, betas, GRS, p -value and the two Sharpe ratios

Interpretation

- ▣ The same test rejects the CAPM on the 25 size $\times$ B/M portfolios (GRS = 4.20). Why is it silent here?

Exercise B2: Solution (1/2) [Solved]

Solution (1/2)

- ▣ (a) Alphas: XLU +3.4%, XLE +2.4%, XLV +1.8%, XLP +1.7%, XLF -2.3% a year; the other four lie within $\pm 0.6\%$
- ▣ Betas: XLU 0.40, XLP 0.45, XLV 0.67, XLE 0.92, XLI 1.04, XLB 1.05, XLF 1.07, XLY 1.07, XLK 1.28
- ▣ (b) $GRS = 0.53$, $p = 0.85$
- ▣ (c) Monthly Sharpe ratios: market 0.146, market plus sectors 0.191 (annual 0.51 and 0.66)
- ▣ Identity: $W = \frac{321}{9} \times \frac{0.191^2 - 0.146^2}{1 + 0.146^2} = 35.67 \times 0.0150 = 0.53$
- ▣ (d) $p = 0.85 > 0.05$: do not reject H_0 : all $\alpha_i = 0$

Exercise B2: Solution (2/2) [Solved]

Solution (2/2)

- Interpretation: the test has little power on these assets
- Sectors mix small and big, value and growth stocks, so they do not spread the characteristics whose premia the CAPM misses; their alphas are small and similar
- The Sharpe-ratio gain from the sectors is small ($0.51 \rightarrow 0.66$ a year), against $0.47 \rightarrow 1.40$ for the 25 portfolios
- Not rejecting is not accepting the CAPM: choose test assets that spread the characteristics suspected to be priced

Exercise B3: PCA and the Number of Factors [Solved]

Task

- **Data:** 11 sector ETFs (the 9 above plus XLRE and XLC), daily log returns on common days, 20 June 2018 – 18 September 2026 ($T = 2073$); SPY for comparison
- (a) Standardise the returns; compute the correlation matrix, its eigenvalues and eigenvectors (sign chosen so that the average loading is positive)
- (b) Report the variance shares of PC1–PC3 and the loadings of PC1 and PC2
- (c) Correlate the PC1 scores with the SPY returns
- (d) Choose the number of factors with Kaiser ($\lambda_k > 1$), Bai & Ng (2002) (IC_{p2}) and the eigenvalue ratio of Ahn & Horenstein (2013), with $k_{\max} = 5$
- **Report:** shares, loadings, the correlation and the three estimates of k

Interpretation

- PC2 opposes utilities and staples to technology: would it earn a premium? What would you need to test it?

Exercise B3: Solution (1/2) [Solved]

Solution (1/2)

- (a) Eigenvalues: 7.26, 1.06, 0.82, 0.40, 0.32, ...
- (b) Shares: PC1 66.0%, PC2 9.7%, PC3 7.4%; the first three explain 83.1%
- PC1 loadings are all positive (0.24–0.34); PC2: XLU +0.55, XLP +0.42, XLRE +0.31 against XLK −0.40, XLY −0.33: defensive versus growth
- (c) $\text{corr}(\text{PC1}, \text{SPY}) = 0.954$: the first latent factor is the market

Exercise B3: Solution (2/2) [Solved]

Solution (2/2)

- (d) Kaiser: 2 (only 7.26 and 1.06 exceed 1)
- Bai–Ng: IC_{p2} is minimised at $k = 5 = k_{\max}$; the penalty $\frac{N+T}{NT} \ln N$ is small when $N = 11$
- Ahn–Horenstein: ratios 6.82, 1.30, 2.05, 1.26, 1.15, maximum at $k = 1$
- The estimators disagree: with $N = 11$ the large- N asymptotics do not apply, so report the range, not a single k
- Interpretation: PCA explains covariance, not average returns. To test a premium, run Fama–MacBeth with the PC2 loadings as betas on many assets, or sort assets on PC2 loadings into a long–short portfolio and test its mean with Newey–West errors

Exercise B4: CAPM Betas of BVB Blue Chips [Proposed]

Solution discussed in class

Task

- **Data:** 11 BVB blue chips (TLV, SNP, BRD, TGN, FP, SNG, H2O, SNN, EL, DIGI, TEL; files <SYMBOL>.R0.csv) and BET.csv; daily log returns on common days from 6 January 2015 or from listing; R_f treated as constant, so it cancels in the slope
- (a) Remove the days on which any $|r_t| > 0.5$ (unadjusted corporate events) and report how many were removed
- (b) Estimate each beta by OLS on BET, with Newey–West errors (lag rule) and R^2
- (c) Compute the empirical-Bayes Vasicek betas of A3, with the 11 stocks as the cross-section
- (d) Compute Dimson betas (Dimson (1979)): regress $r_{i,t}$ on $r_{\text{BET},t-1}$, $r_{\text{BET},t}$, $r_{\text{BET},t+1}$ and sum the three slopes
- **Report:** a table with $\hat{\beta}$, standard error, R^2 , Vasicek and Dimson betas, and a chart of the betas with 95% intervals. **Model:** B1, A3

Interpretation

- How does asynchronous trading of less liquid stocks bias daily betas, and what does the Dimson column show?

Exercise B5: Fama–MacBeth on ETFs [Proposed]

Solution discussed in class

Task

- ▣ **Data:** 16 test assets, the 9 sector ETFs and 7 factor ETFs (MTUM, VVALUE, QUAL, USMV, IWM, IWD, IWF); monthly excess returns August 2013 – July 2026 ($T = 156$); factors Mkt-RF, SMB, HML
- ▣ (a) Pass 1: estimate full-sample time-series betas of each ETF on the three factors
- ▣ (b) Pass 2: run 156 monthly cross-sectional OLS regressions of returns on $(1, \hat{\beta}_i)$ and average the slopes
- ▣ (c) Compute Fama–MacBeth, Shanken, $\text{Var}_{\text{sh}} = (1 + c)(\text{Var}_{FM} - \Sigma_f / T) + \Sigma_f / T$, and Kan–Robotti–Shanken errors (Kan, Robotti & Shanken (2013))
- ▣ (d) Compare each $\hat{\lambda}$ with the time-series mean of its factor and test $\lambda = 0$ at 5%
- ▣ **Report:** a table with $\hat{\lambda}$, the three standard errors and the factor means. **Model:** B2 and Lecture 3, Section 5

Interpretation

- ▣ Is a non-significant premium here evidence that size and value are not priced?

Exercise B6: Do Factor ETFs Have Alpha? [Proposed]

Solution discussed in class

Task

- **Data:** adjusted prices of MTUM and QUAL (data/market); daily FF5 (Fama & French (2015)) and MOM (Carhart (1997)) factors; each ETF on its own trading calendar
- (a) Build daily excess returns, keeping only returns between consecutive factor trading days
- (b) Estimate the six-factor regression and annualise α (multiply by 252)
- (c) Compute the Newey–West standard error (10 lags) and the t -statistic of α ; test $\alpha = 0$ at 5%
- (d) Build a 95% moving-block bootstrap interval for α : blocks of 20 days, 1000 draws
- **Report:** n , α , Newey–West error, t and the bootstrap interval for each ETF. **Model:** B1

Interpretation

- Why must a 3-day ETF return never be regressed on a 1-day factor return?

Exercise B7: Multiple Testing of 25 Alphas [Proposed]

Solution discussed in class

Task

- **Data:** the 25 size $\times$ B/M portfolios and the FF3 factors (Fama & French (1993)), monthly excess returns July 1963 – July 2026 ($T = 757$)
- (a) Estimate the 25 FF3 time-series regressions; compute the Newey–West t -statistic of each alpha (lag rule) and the two-sided p -value from the Normal distribution
- (b) Count the discoveries at 5%: naive, Bonferroni, Holm and Benjamini–Hochberg, and the number with $|t| > 3$ (Harvey, Liu & Zhu (2016))
- (c) Identify the portfolio with the largest $|t|$ and report its alpha
- **Report:** the five counts and the largest alpha with its t . **Model:** A5

Interpretation

- The 25 tests are strongly correlated: which methods remain valid, and which test uses the correlation directly?



Exercise B8: PCA of BVB Blue Chips [Proposed]

Solution discussed in class

Task

- ▣ **Data:** 10 blue chips (H2O excluded: listed in 2023) and BET; daily log returns on common days from 7 June 2017; remove days with any $|r_t| > 0.5$
- ▣ (a) Standardise the returns; compute the eigenvalues, the variance shares and the PC1 loadings
- ▣ (b) Correlate the PC1 scores with the BET returns
- ▣ (c) Choose the number of factors with Kaiser, Bai–Ng (IC_{p2}) and Ahn–Horenstein, with $k_{\max} = 5$
- ▣ (d) Compare the PC1 share and the number of factors with the US sectors (B3)
- ▣ **Report:** shares, loadings, the correlation and the three estimates of k . **Model:** B3

Interpretation

- ▣ What does a weaker first factor imply for diversification within the BVB?

Exercise B9: A Lewellen–Nagel–Shanken Check (1/2) [Proposed]

Solution discussed in class

Task (1/2)

- **Data:** 25 size $\times$ B/M portfolios and 30 industry portfolios (Kenneth French Data Library), monthly excess returns July 1963 – July 2026 ($T = 757$); models CAPM, FF3 and FF5 (Lewellen, Nagel & Shanken (2010))
- (a) Estimate pass-1 betas; regress the mean returns on $(1, \hat{\beta}_i)$ by OLS and compute the OLS R^2
- (b) Repeat by GLS with weight $\hat{V}^{-1}$ (sample covariance of the returns): $R_{GLS}^2 = 1 - \mathbf{e}^\top \hat{V}^{-1} \mathbf{e} / (\boldsymbol{\mu} - \bar{\boldsymbol{\mu}}_{GLS})^\top \hat{V}^{-1} (\boldsymbol{\mu} - \bar{\boldsymbol{\mu}}_{GLS})$

Exercise B9: A Lewellen–Nagel–Shanken Check (2/2) [Proposed]

Solution discussed in class

Task (2/2)

- (c) Build 95% intervals by an i.i.d. bootstrap over months (500 draws), re-estimating both passes in each draw
- (d) Repeat (a)–(c) on the 55 assets; for FF3 also impose $\lambda = \mathbb{E}[f]$ with no intercept
- **Report:** a table of OLS and GLS R^2 with intervals for the 3 models and 2 sets of assets. **Model:** B2 and Lecture 3, Section 5

Interpretation

- Does the FF3 R^2 survive once industries are added?

 MFM_ch3_two_pass_inference

Exercise B10: Comparing Models by Sharpe Ratios (1/2) [Proposed]

Solution discussed in class

Task (1/2)

- **Data:** monthly factors Mkt-RF, SMB, HML, RMW, CMA, MOM ([Kenneth French Data Library](#)), July 1963 – July 2026 and January 2000 – July 2026; models CAPM, FF3, Carhart (FF3 + MOM), FF5 and FF5 + MOM ([Barillas & Shanken \(2018\)](#))
- (a) Compute each model's maximum squared Sharpe ratio $SR^2 = \mu_f^\top \Sigma_f^{-1} \mu_f$, annualise it (multiply by 12) and report its square root
- (b) Build 95% percentile intervals for the SR^2 differences FF3 – CAPM, FF5 – FF3, FF5 + MOM – FF5 (nested) and FF5 – Carhart (non-nested), with a stationary bootstrap ([Politis & Romano \(1994\)](#)), mean block 6 months, 2000 draws

Exercise B10: Comparing Models by Sharpe Ratios (2/2) [Proposed]

Solution discussed in class

Task (2/2)

- (c) Regress each of the six factors on the other five (spanning regressions) and report the Newey–West t of the intercept
- (d) Decide which factors are redundant at 5% in each period
- **Report:** the Sharpe-ratio table, the four intervals and the six spanning t per period. **Model:** B1

Interpretation

- Why is a bootstrap interval useless for nested models? Which factors are redundant after 2000?

 MFM_ch3_two_pass_inference

Exercise B11: GRS without Normality (1/2) [Proposed]

Solution discussed in class

Task (1/2)

- **Data:** 25 size $\times$ B/M portfolios, CAPM with Mkt-RF, monthly July 1963 – July 2026 ($T = 757$, $N = 25$)
- (a) Estimate the 25 regressions; compute the kurtosis of each residual series and the average kurtosis
- (b) Compute GRS with the maximum-likelihood $\hat{\Sigma}$ and its p -value from $F(25, 731)$

Exercise B11: GRS without Normality (2/2) [Proposed]

Solution discussed in class

Task (2/2)

- (c) Residual bootstrap under H_0 : resample whole rows of residuals (keeping their cross-correlation), rebuild $R_t^* = \hat{\beta}f_t + \epsilon_t^*$, recompute GRS 2000 times; report the p -value and the 95% bootstrap critical value
- (d) GMM Wald test on the moments $\epsilon_t \otimes (1, f_t)$: $W = \hat{\alpha}^\top \hat{V}_\alpha^{-1} \hat{\alpha} \sim \chi_{25}^2$, with White and with Newey–West (6 lags) covariances
- (e) Explain why GRS cannot be computed for the 100 size $\times$ B/M portfolios over August 2018 – July 2026 ($T = 96$)
- **Report:** the kurtosis, the four p -values and the critical values. **Model:** B2

Interpretation

- Is the rejection of the CAPM an artefact of assuming Normal residuals?

C1: Do Factor Premia Exist on the Bucharest Stock Exchange? [Proposed]

Solution discussed in class

Task

- **Question:** do momentum (Jegadeesh & Titman (1993)) and low beta (Frazzini & Pedersen (2014)) earn premia among BVB blue chips?
- **Data:** month-end adjusted prices of the 10 blue chips of B8 and the BET index, from December 2015; monthly simple returns; months with $|r| > 0.5$ are treated as data errors
- (a) Build the 12–1 momentum signal (return from month $t - 12$ to $t - 2$), known at the start of month t
- (b) Each month, go long the 3 stocks with the highest signal and short the 3 with the lowest, with equal weights
- (c) Low beta: estimate 36-month rolling betas on BET (at least 18 months of data); go long the 3 lowest-beta and short the 3 highest-beta stocks
- (d) Report the mean, the volatility and the Newey–West t (3 lags) of each long–short return; compare t with 1.96 and with the $|t| > 3$ hurdle of the factor zoo
- (e) Discuss the number of stocks, short-selling constraints, transaction costs and liquidity; name public sources of book equity and shares outstanding for size and value
- **Deliverable:** a Quantlet folder in the team repository and one slide with the conclusion and its main limitation

C2: Critique an AI Answer (1/2) [Proposed]

Solution discussed in class

- **The prompt** a student sent to an AI (Artificial Intelligence) assistant: “P holds cleaned month-end adjusted prices of the BVB blue chips of C1, one column per stock. Build a 12-month momentum long–short portfolio, top 3 minus bottom 3 with equal weights, and report its annual mean and volatility”
- **The answer:** the code below, followed by the conclusion

```
R      = np.log(P).diff()                # monthly returns
mom    = P.pct_change(12)                # past 12-month return
ls     = {}
for t in R.index[13:]:
    s = mom.loc[t].dropna().sort_values() # rank the stocks
    ls[t] = R.loc[t, s.index[-3:]].mean() - R.loc[t, s.index[:3]].mean()
ls = pd.Series(ls)
print(12 * ls.mean(), np.sqrt(12) * ls.std()) # annual mean, volatility
```

- **Conclusion of the AI:** “Momentum earns about 49% a year with a volatility of 16%. This agrees with [Fama & French \(2012\)](#), who document momentum on the Bucharest Stock Exchange.”

C2: Critique an AI Answer (2/2) [Proposed]

Solution discussed in class

Task

- ▣ (a) Find the errors: there are two or three; the rest of the answer is correct
- ▣ (b) For each error, say how you would detect it: a test, a published number, the lecture or the paper itself
- ▣ (c) Give the corrected code and the numbers it produces
- ▣ (d) Log each error as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, the correct version
- ▣ Model: C1 and Lecture 3

Why this exercise

- ▣ AI code usually runs without an error message: the mistakes are in the finance, not in the syntax
- ▣ Course rule: you may use AI, declare it in `AI_USE.md` and log at least three errors you caught in `AI_ERRORS.md`

Common Pitfalls

Data

- ▣ Unadjusted corporate events (spin-offs, capital returns) create fake returns
- ▣ Regressing multi-day asset returns on one-day factors
- ▣ Forgetting that French data are in percent
- ▣ Using signals that were not known at the portfolio date

Statistics

- ▣ Classical errors for betas under volatility clustering
- ▣ Fama–MacBeth errors without the Shanken or misspecification correction
- ▣ Trusting a high cross-sectional R^2 on 25 portfolios
- ▣ $|t| > 1.96$ after trying many factors
- ▣ Reading “not rejected” as “true”

- ▣ Rule: state the sample, the frequency, the market proxy and the risk-free rate of every regression

Take-Home Assignment (Team Project)

- ▣ Choose **five** assets: at least two BVB stocks, one US sector ETF and one factor ETF
- ▣ For each asset
 - ▶ CAPM beta with Newey–West errors, a rolling beta and its Vasicek-adjusted value
 - ▶ For US assets: FF5 + MOM loadings and alpha with a bootstrap interval
- ▣ Jointly: GRS test and a PCA of the five assets; compare with Lecture 3
- ▣ Deliverables: a GitHub repository with a notebook that runs top-to-bottom and a 2-page summary
- ▣ Textbook for PCA: [Härdle & Simar \(2019\)](#)
- ▣ Log every AI error you catch, in this seminar or in the project, in `AI_ERRORS.md`

Further Study on Quantinar

Advanced courses

- ▣  Generalized Method of Moments
- ▣  Multivariate Statistical Analysis

Code

- ▣ **Notebook:** `chapter3_seminar_notebook.ipynb` (Colab)
- ▣ **Quantlets:** github.com/danpele/MFM/tree/main/Quantlets/Ch_03

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